

For example, we assume the number of the independent variables to find optimum word-length N is two and the base position is

We have generalized the minimum number of experiments of the full search in N-dimension as

where $d = dw_1 + dw_2 + \dots + dw_N$. (7)

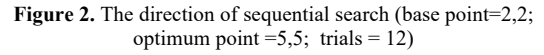
2.3.2 Sequential Search (SS)

The principles of sequential search in n dimensions can be summarized in the following six steps:

- In step 3, the direction of search is chosen for the best performance in accordance with

where

For example, we assume the same condition as the example of the full search for comparison. In Fig. 2, starting from base point (2,2), we simulate (2,3) and (3,2) from the direction of sequential search in step 3. If the performance of (3,2) is better than that of (2,3), move into (3,2) that is a new word-length set. Simulation would be repeated until satisfying the desired performance.


$$E_{SS}^N = N \cdot (dw_1 + dw_2 + \dots + dw_N). \quad (10)$$

2.3.3 Preplanned Search (PS)

The principles of preplanned search in n dimensions can be summarized in the following four steps:

- In step 2, the direction of preplanned search is chosen in accordance with maximum derivative of an independent performance:

$$\vec{\xi}_j = \begin{cases} (1,0,0,...,0) & \text{if } m_j = \frac{\partial p}{\partial w_1} \\ (0,1,0,...,0) & \text{if } m_j = \frac{\partial p}{\partial w_2} \\ (0,0,1,...,0) & \text{if } m_j = \frac{\partial p}{\partial w_3} \\ \\ (0,0,0,...,1) & \text{if } m_j = \frac{\partial p}{\partial w_N} \end{cases} \quad (11)$$

where $m_j = \max(\frac{\partial y}{\partial w_1}, \frac{\partial y}{\partial w_2}, \dots, \frac{\partial y}{\partial w_n})$ (12)

For example, we assume the same condition as the full search. In Fig. 3, starting from base point (2,2), we preplan the directions of steepest ascent by comparing the gradients of the independent performances at one dimension. If the gradient, calculated from the one-variable simulation in section 2.2, at w_1 of 2 is larger than that at w_2 of 2, the next feasible location is (3,2). If the gradient at w_1 of 3 is smaller than that at w_2 of 2, the next feasible location is (3,3). The preplanned path would be (2,2), (3,2), (3,3), (4,3)... After scheduling the feasible positions, evaluate the performance of these positions, until the value of the performance satisfy the desired accuracy.

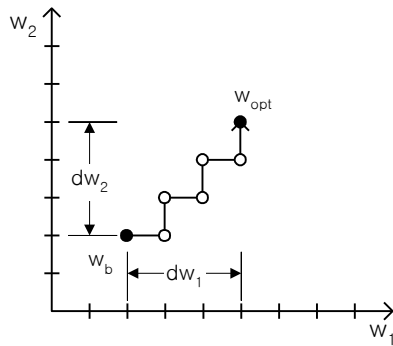


Figure 3. The direction of preplanned search (base point=2,2;
optimum point =5,5; trials = 6)

We generalized the trials of the preplanned search in N-dimension as

$$E_{PS}^N = dw_1 + dw_2 + \dots + dw_N \quad (13)$$

In this example, the trials are 6 from (13) and 6 from by count Fig. 3.

3. SIMULATION

We applied this optimization to determine the optimum word-length for CDMA demodulator.

3.1 CDMA Demodulator

Typical demodulators are implemented with an analog block in front of an analog digital converter (ADC) block as shown in fig.4 (a). As the speed of the ADC is increasing, analog parts have been replaced with digital parts in communication systems [5]. We replaced the analog demodulator with the digital demodulator as shown in Fig. 4 (b).

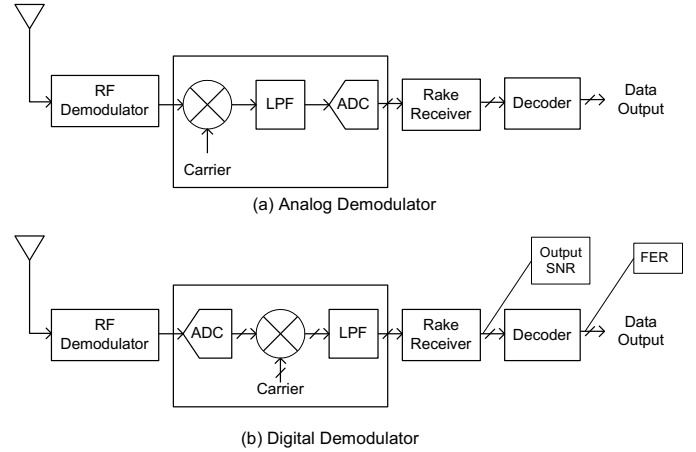


Figure 4. Analog and digital demodulator on CDMA receiver and performance measurement

The demodulator converts modulated signals into baseband signals. In the digital demodulator block of Fig. 5, the sampled data at the ADC are multiplied by a carrier signal down to the baseband signal and removed the out-of-band signal by the low pass filter (LPF). The variables in the digital demodulator are as below [6]:

- B_i : input word-length
- B_c : carrier word-length
- B_m : multiplier output word-length
- B_f : filter output word-length
- B_{fc} : filter coefficient word-length.

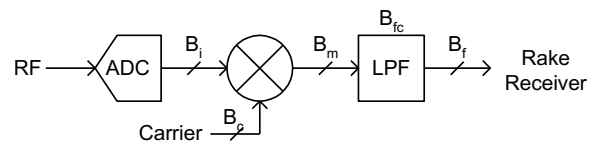


Figure 5. A digital demodulator block.

We have used the output SNR instead of frame error rate (FER) which is a general measurement to evaluate CDMA systems, for measurement of the performance, because direct measurement of FER requires at least 10^5 data during simulation. The required output SNR is over 0.8 dB or FER is under 0.03 [7].

Commonly the base position of $(B_i, B_c, B_m, B_f, B_{fc})$ over 0.8 dB is (4,3,4,5,7), acquired from the one-variable simulations shown as Fig 6. We assume cost per bit is similar. In the FS, the next positions are exhaustively (5,3,4,5,7), (4,4,4,5,7), (4,3,5,5,7), (4,3,4,6,7), (4,3,4,5,8), (5,4,4,5,7) and etc. The search is continued until the performance is over the specific desired requirement.

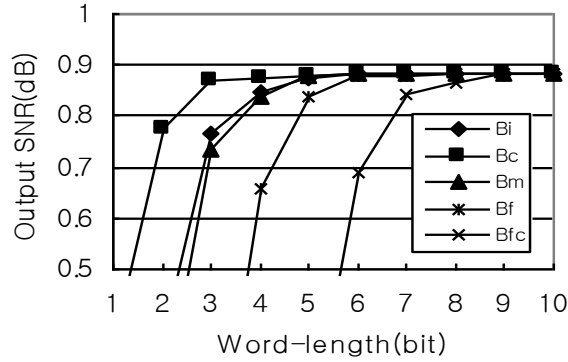


Figure 6. Result of independent one-variable simulations on a CDMA demodulator

In the SS, the next position is one of (5,2,5,6,6), (4,3,5,6,6), (4,2,7,6,6), (4,2,6,7,6) and (4,2,5,6,7). The next position would be the set having the maximum performance of them. From table 1, (4,3,4,6,7) is the next location because it has the maximum performance. Move into the new position and continue to search until the performance is over the specific desired requirement, over output SNR of 0.8. The optimum region is (5,3,6,6,7). Distance between the base and the optimum position is 4.

In the PS, search path is estimated from the sensitivity of each one-variable simulation shown in Fig. 6. Starting from the base position (4,3,4,5,7), the first expected position is (4,3,4,6,7) because the B_f has the greatest derivative among each word-length at the base position from Fig. 5. The preplanned search positions are (4,3,4,5,7), (4,3,4,6,7), (4,3,4,6,8), (4,3,5,6,8), (4,4,5,6,8), etc. Move into the preplanned positions until the performance is over the specific desired requirement.

3.2 COMPARISON

The three search methods are compared with the trials from (6), (10), (13), as shown in table 2. The FS is inefficient to find the optimum word-length when the variables for optimization are numerous and the distance between base and optimum position is long. The SS is easy to be applied to digital systems and has fewer numbers of experiments. The PS, which needs the calculation of the gradients of the performance at each variable, has the least number of experiments among search methods. The SS and the PS reduced the trials 64% and 92%, respectively, compared to the FS in word-length optimization for the CDMA demodulator.

4. CONCLUSION

In this paper, numerical search method to optimize the word-length for digital implementations are explained. We consider the FS and propose SS and PS. We have generalized the trials as (6), (10), (13). The FS is inefficient with regard to the trials. The SS requires fewer experiments. The PS needs the least number of experiments when the variables for optimization are numerous and the distance between the base and the optimum position is long. We have applied those methods to a CDMA demodulator design. The simulation results show the SS and PS reduced the number of trials 64%, 92% respectively compared to full search finding the optimum word-length in the CDMA demodulator. In

the future, an adaptive step size for faster search would be considered.

Table 1. Sequence of the Sequential Search for CDMA demodulator (Traffic Ch. Rate Set 1 in AWGN, Input SNR = -17.3 dB, $E_b/N_t = 3.8$, Rate= 9600 bps, Desired performance: Output SNR>0.8, FER< 0.03)

Step	Bi	Bc	Bm	Bf	Bfc	Output SNR	FER	Result
1, 2	4	3	4	5	7	0.711	0.0384	Fail
	5	3	4	5	7	0.735	-	-
	4	4	4	5	7	0.694	-	-
3, 4	4	3	5	5	7	0.712	-	-
	4	3	4	6	7	0.759	-	Max
	4	3	4	5	8	0.704	-	-
5, 6	4	3	4	6	7	0.759	0.035	Fail
	5	3	4	6	7	0.763	-	-
	4	4	4	6	7	0.722	-	-
3, 4	4	3	5	6	7	0.773	-	Max
	4	3	4	7	7	0.751	-	-
	4	3	4	6	8	0.749	-	-
5, 6	4	3	5	6	7	0.773	0.0342	Fail
	6	3	5	6	7	0.798	-	-
	5	4	5	6	7	0.802	-	-
3, 4	5	3	6	6	7	0.805	-	Max
	5	3	5	7	7	0.803	-	-
	5	3	5	6	8	0.798	-	-
5, 6	5	3	6	6	7	0.805	0.0292	Pass

Table 2. Comparison of full, sequential and preplanned searches (N=5, $d=dw_1+dw_2+dw_3+dw_4+dw_5=4$)

Type	Equation for Number of Experiments from (6), (10), (13)	Trials
FS	$(d+4)(d+3)(d+2)(d+1) \cdot d/5!$	56
SS	$5 \cdot d$	20
PS	d	4

5. References

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